National
Qualifications
SPECIMEN ONLY

S844/75/02

Applications of Mathematics
Paper 2

Marking Instructions

These marking instructions have been provided to show how SQA would mark this specimen question paper.

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General marking principles for National 5 Applications of Mathematics

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

For each question, the marking instructions are generally in two sections:

generic scheme — this indicates why each mark is awarded
illustrative scheme — this covers methods which are commonly seen throughout the marking

In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.

(c) One mark is available for each •. There are no half marks.

(d) If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.

(e) Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.

(f) Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.

(g) If an error is trivial, casual or insignificant, for example 6 x 6 = 12, candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.

(h) If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$
$$x - 4x + 3 = 0$$
$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$
$$x - 4x + 3 = 0$$
$$(x-3)(x-1) = 0$$
$$x = 1 \text{ or } 3$$

(i) **Horizontal/vertical marking**
If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

	$\bullet^5$	$\bullet^6$
$\bullet^5$	$x = 2$	$x = -4$
$\bullet^6$	$y = 5$	$y = -7$

Horizontal: $\bullet^5$ $x = 2$ and $x = -4$
$\bullet^6$ $y = 5$ and $y = -7$

Vertical: $\bullet^5$ $x = 2$ and $y = 5$
$\bullet^6$ $x = -4$ and $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$\frac{15}{12}$ must be simplified to $\frac{5}{4}$ or $1\frac{1}{4}$

$\frac{43}{1}$ must be simplified to 43

$\frac{15}{0.3}$ must be simplified to 50

$\frac{4/5}{3}$ must be simplified to $\frac{4}{15}$

$\sqrt{64}$ must be simplified to 8*

*The square root of perfect squares up to and including 144 must be known.

(k) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example

  $(x^3+2x^2+3x+2)(2x+1)$ written as

  $(x^3+2x^2+3x+2)\times 2x+1$

  $=2x^4+5x^3+8x^2+7x+2$

  gains full credit

- repeated error within a question, but not between questions or papers

(l) In any 'Show that…' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

(m) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.

(n) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.

(o) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

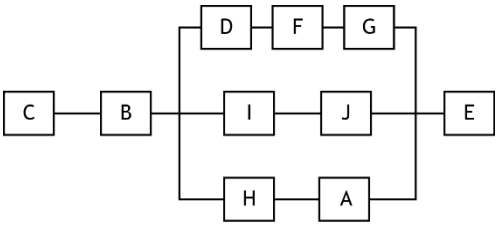
Strategy 1 attempt 1 is worth 3 marks.	Strategy 2 attempt 1 is worth 1 mark.
Strategy 1 attempt 2 is worth 4 marks.	Strategy 2 attempt 2 is worth 5 marks.
From the attempts using strategy 1, the resultant mark would be 3.	From the attempts using strategy 2, the resultant mark would be 1.

In this case, award 3 marks.

Marking Instructions for each question

Question			Generic scheme	Illustrative scheme	Max mark
1.	(a)		•1 Strategy: identify multiplier •2 Strategy: identify power •3 Process: calculate value •4 Communication: round to 2 significant figures	•1 0.87 •2 …3 •3 921.90(42) •4 920	**4**
	(b)		•5 Strategy: know how to calculate percentage loss •6 Process: calculate percentage	•5 $\frac{450}{1400}\times 100$ •6 32(.1…)	**2**
			Alternative Strategy •5 Strategy: know to use trial and improvement •6 Process: calculate percentage	•5 evidence •6 32	**2**
2.	(a)	(i)	•1 Process: calculate mean	•1 70.5	**1**
		(ii)	•2 Process: calculate $(x-\bar{x})^2$ •3 Strategy/process: calculate $\sum(x-\bar{x})^2$ and substitute into formula •4 Process: calculate standard deviation	•2 2.25, 20.25, 6.25, 6.25, 30.25, 0.25 •3 $\sqrt{\frac{65.5}{6-1}}$ •4 3.62	**3**
			Alternative Strategy •2 Process: calculate $\sum x$ and $\sum x^2$ •3 Strategy/process: substitute into formula •4 Process: calculate standard deviation	•2 423, 29887 •3 $\sqrt{\frac{29887-\frac{423^2}{6}}{6-1}}$ •4 3.62	**3**
	(b)		•5 Communication: comment regarding mean •6 Communication: comment regarding standard deviation	•5 eg on average prices in August were cheaper. •6 eg prices in August were less consistent	**2**

Question			Generic scheme	Illustrative scheme	Max mark
3.			•[1] Strategy: know how to find arc length of quarter or semi-circle •[2] Process: calculate curved edge of one quarter circle or semi-circle •[3] Process: calculate perimeter of swimming pool •[4] Strategy: know how to calculate number of lengths •[5] Process: calculate number of lengths, appropriate rounding and calculate cost	•[1] $\frac{20\pi}{4}$ or $\frac{20\pi}{2}$ •[2] 15.7…or 31.4… •[3] $2\times15.7\ldots+2\times10+2\times36.5=124.4\ldots$ •[4] $(\ldots-2\times1.25)\div3$ •[5] 40.6 leading to $41\times11.49=471.09$	**5**
4.	(a)		•[1] Process: calculate time in hours •[2] Process/communication: calculate the time of arrival	•[1] 0.55 •[2] 08:12 + 33 mins = 08:45	**2**
	(b)		•[3] Strategy/process: convert litres to gallons •[4] Strategy/process: •[5] Strategy/process: convert km to miles •[6] Communication: state conclusion consistent with working	•[3] $5\div4.545=1.1\ldots$ •[4] $1.1\ldots\times67=73.7\ldots$ •[5] $73.7\div1.609=45.8\ldots$ •[6] No, since $45.8<52$	**4**

Question			Generic scheme	Illustrative scheme	Max mark
5.	(a)		•[1] Process: calculate total number of hours •[2] Process: calculate angles •[3] Communication: draw and label pie chart consistent with previous working	•[1] 210 + 96 + 234 = 540 •[2] 210 hours research: $\frac{210}{540} \times 360 = 140$ 96 hours meetings: $\frac{96}{540} \times 360 = 64$ 234 hours writing: $\frac{234}{540} \times 360 = 156$ •[3] diagram consistent with working	**3**
Notes: 1. •[1] and •[2] can be implied in subsequent working 2. •[3] is available if any 2 angles are within tolerance $\pm 1°$ leading to third angle being outwith tolerance 3. •[3] is not available if the three calculated angles do not add to $360°$					
	(b)		•[4] Communication: any 5 in correct sequence •[5] Communication: remaining 4 in correct sequence		**2**
	(c)		•[6] Strategy: know to and starts to calculate the correct two ways of packing •[7] Process: calculate number of boxes for one arrangement •[8] Process/communication: calculate the second arrangement and state maximum number of books	•[6] evidence of the two **correct** ways of packing with the front cover facing upwards •[7,8] $100 \div 12.5 = 8$ $50 \div 19.5 = 2.56...$ $20 \div 2 = 10$ $2 \times 8 \times 10 = 160$ and $100 \div 19.5 = 5.12...$ $50 \div 12.5 = 4$ $20 \div 2 = 10$ $5 \times 4 \times 10 = 200$ Maximum - 200 books	**3**

Question			Generic scheme	Illustrative scheme	Max mark
5.	(d)		•[9] Process: calculate cost of shop A •[10] Process: calculate cost of shop C •[11] Communication: conclusion consistent with working	•[9] 24 •[10] 22 •[11] Shop C	**3**
			Alternative Strategy •[9] Process: calculate discount for 1 shop •[10] Process: calculate discount for other two shops •[11] Communication: conclusion consistent with working	•[9] 6 or 7.01 or 8 •[10] remaining two •[11] Shop C	
6.	(a)		•[1] Strategy: know how to calculate minimum price •[2] Process: calculations completed with appropriate rounding	•[1] Evidence •[2] $(38.20+20)\div(24\times10)=0.2425$ leading to 0.25	**2**
	(b)	(i)	•[3] Process: calculate scale distances •[4] Process/communication: correct bearing measured and correct length drawn •[5] Process/communication: correct bearing measured and correct length drawn	•[3] $140\div40=3.5\text{cm}$ $252\div40=6.3\text{cm}$ •[4] Bearing of $055^\circ(\pm1^\circ)$ measured correctly and 3.5 cm(±0.1 cm) correctly drawn •[5] Bearing of $170^\circ(\pm1)$ measured correctly and 6.3 cm(±0.1 cm) correctly drawn	**3**
		(ii)	•[6] Process: bearing consistent with diagram •[7] Process: distance consistent with diagram	•[6] evidence •[7] evidence	**2**

<table>
<tr><th colspan="3">Question</th><th>Generic scheme</th><th>Illustrative scheme</th><th>Max mark</th></tr>
<tr><td>6.</td><td>(c)</td><td></td><td>•8 Process/communication: calculate lucky dip probability

•9 Strategy/process: find all combinations for two dice

•10 Process: find the number of combinations 9 or more

•11 Process/communication: calculate probability and compare</td><td>•8 $\frac{30}{150}$ (or 0.2)

•9 Evidence of 36 combinations

•10 10

•11 $\frac{10}{36}$ (or 0.278)
since $0.2 < 0.278$ dice game has a better chance</td><td>4</td></tr>
<tr><td>7.</td><td>(a)</td><td></td><td>•1 Strategy/Process: change to consistent units

•2 Process: calculate volume of conditioner</td><td>•1 20 (l) or 14 000(ml)

•2 0.0035 (l) or 3.5 (ml)</td><td>2</td></tr>
<tr><td></td><td>(b)</td><td></td><td>•3 Strategy: substitute correctly into cylinder formula

•4 Process: calculate volume of cylinder

•5 Strategy/Process: calculate volume of cuboid with height 40cm

•6 Strategy/Process: calculate volume of water</td><td>•3 $\pi \times 5^2 \times 8$

•4 628.318…

•5 36 000

•6 35 371.6…</td><td>4</td></tr>
</table>

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Question			Generic scheme	Illustrative scheme	Max mark
7.	(c)		•[7] Strategy/communication: correct substitution into Pythagoras' theorem •[8] Process: calculate length of diameter •[9] Process: calculate area of table top •[10] Process/communication: convert to square metres	•[7] $30^2 + 30^2$ •[8] 42.426… •[9] 1413.7… •[10] 0.14137…	**4**

[END OF SPECIMEN MARKING INSTRUCTIONS]